Evaluating Orthogonal Projections for Interpreting Vector Embedding Space

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Code Available

Motivation

Semantic classification is useful for a variety of tasks, including misinformation detection and content retrieval. However, most frameworks for semantic classification require:

1. Large and diverse corpuses of labeled training data
2. Model finetuning or training supervised classifiers

These methods are **computationally expensive** and **lack interpretability.**

Question

Can semantic distinctions be identified from vector embeddings through a process that leverages simple human insight to substitute large datasets and computational cost?

Method: Orthogonal Projection

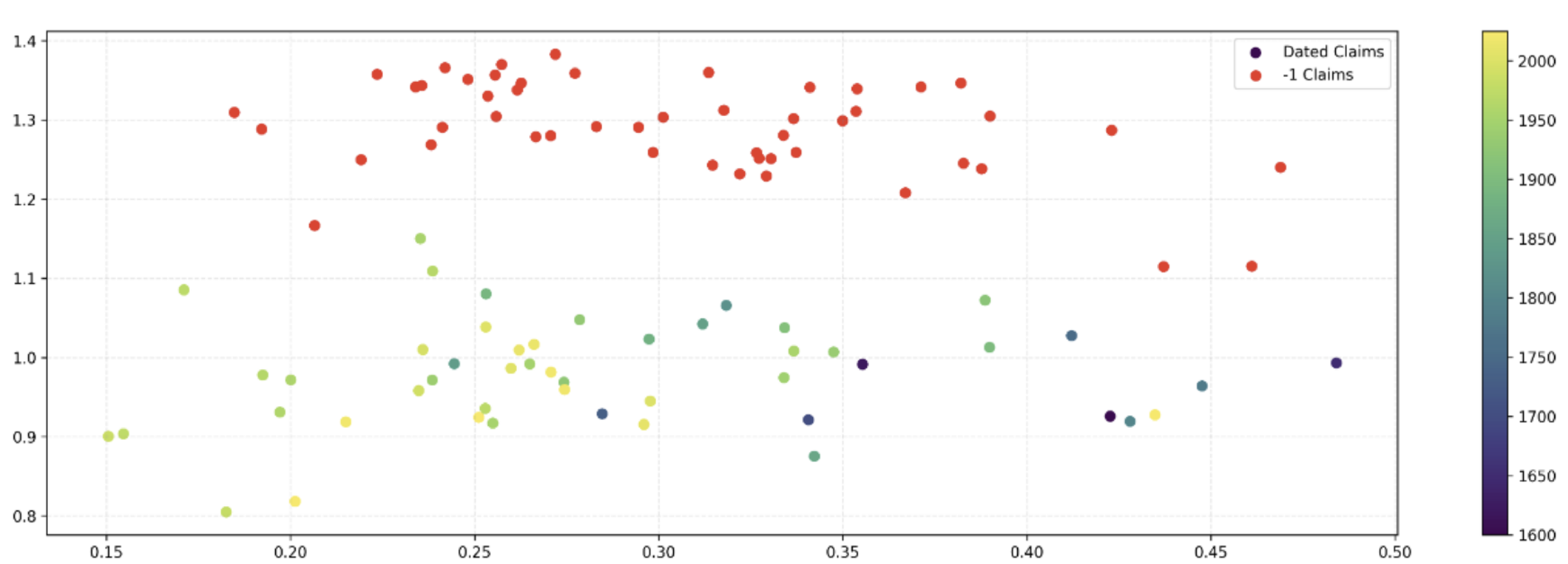
This method uses human insight on the metric of classification to define *reference point statements*, which can be translated into *reference vector embeddings*

Given two reference embeddings r_A, r_B and any Candidate embedding e, we compute:

$$d_1 = \|\mathbf{e}_i - \mathbf{r}_\mathrm{A}\|_2 \qquad d_2 = \|\mathbf{e}_i - \mathbf{r}_\mathrm{B}\|_2 \qquad x_i = \frac{d_1^2 - d_2^2 + d_{12}^2}{2 \cdot d_{12}}$$

This orthogonal projection gives us:

1. Scalar projection *x*: position along reference axis (stance within human-defined topic)
2. Orthogonal distance *y*: deviation from axis (relevance to human-defined topic)



$r_A = 0$: *Smoking is harmful to health* → $r_B = 100$: *Smoking is good for health*

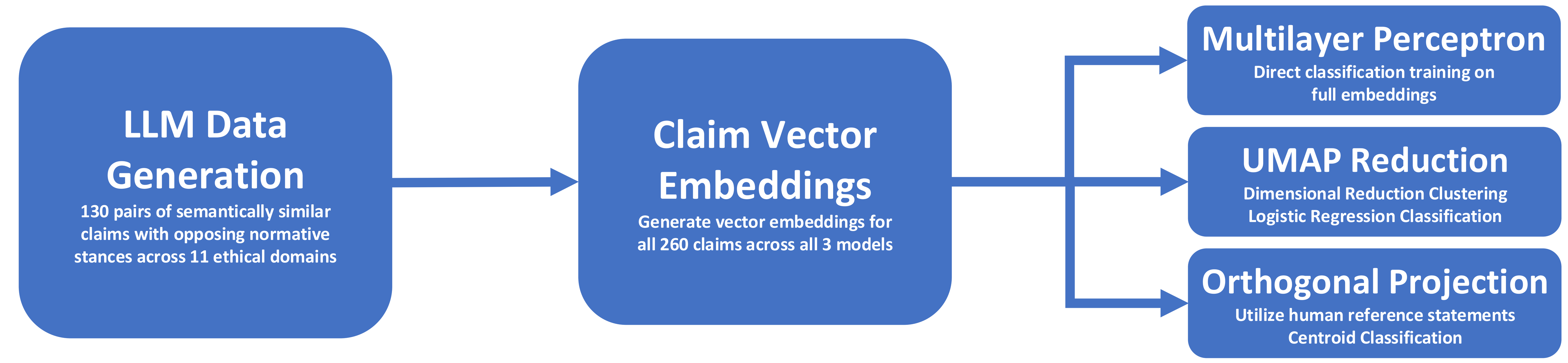
Evaluating Multiple Embedding Models

voyage
voyage-2
(1024 Embedding Dimensions)

OpenAI
text-embedding-3-small
(1536 Embedding Dimensions)

mixedbread-ai
mxbai-embed-large
(768 Embedding Dimensions)

Experimental Procedure



Evaluation results

1. Comparison to MLP highlights non-trivial information loss in orthogonal projection
2. Uninformed UMAP focuses on cross-topic differences, not the ethical differences
3. Orthogonal projection viably captures discriminative information with low computational costs and no training data requirement

	LR (2D Proj.)		LR (UMAP 2D)		MLP (Full Emb.)		Centroid (2D)	
Model	**Acc**	**AUC**	**Acc**	**AUC**	**Acc**	**AUC**	**Acc**	**AUC**
OpenAI Small	0.662	0.720	0.554	0.527	**0.908**	**0.968**	0.631	—
mxbai-embed	**0.738**	**0.778**	0.477	0.481	**0.908**	0.954	0.662	—
Voyage Lite	0.600	0.713	0.446	0.459	0.892	0.925	0.692	—
Mean	*0.667*	*0.737*	*0.492*	*0.489*	*0.903*	*0.949*	*0.662*	—